

5 The photograph shows a camel.



(Source: © Shengyong Li/Shutterstock)

Camels are an important livestock species. They are adapted to living in hot, dry environments.

(a) How many of the following adaptations enable a camel to reduce its water loss?

- production of very concentrated urine
- high glomerular filtration rate
- long loop of Henle
- reduced release of ADH

(1)

☐ **A** 0

☐ **B** 1

☐ **C** 2

☐ **D** 3

(b) The table shows the water use of five species of animal.

Species	Water use / $\text{cm}^3 \text{kg}^{-1} \text{day}^{-1}$	Mean mass / kg
buffalo	150	697
camel	57	850
goat	136	80
sheep	95	110
zebu	120	195

- (i) Calculate the difference in the volume of water use in a day by a buffalo and a camel.

Give your answer in dm^3 to two significant figures.

(2)

Answer dm^3

- (ii) Suggest two reasons why there is no correlation between water use and mean mass for the species given in the table.

(2)

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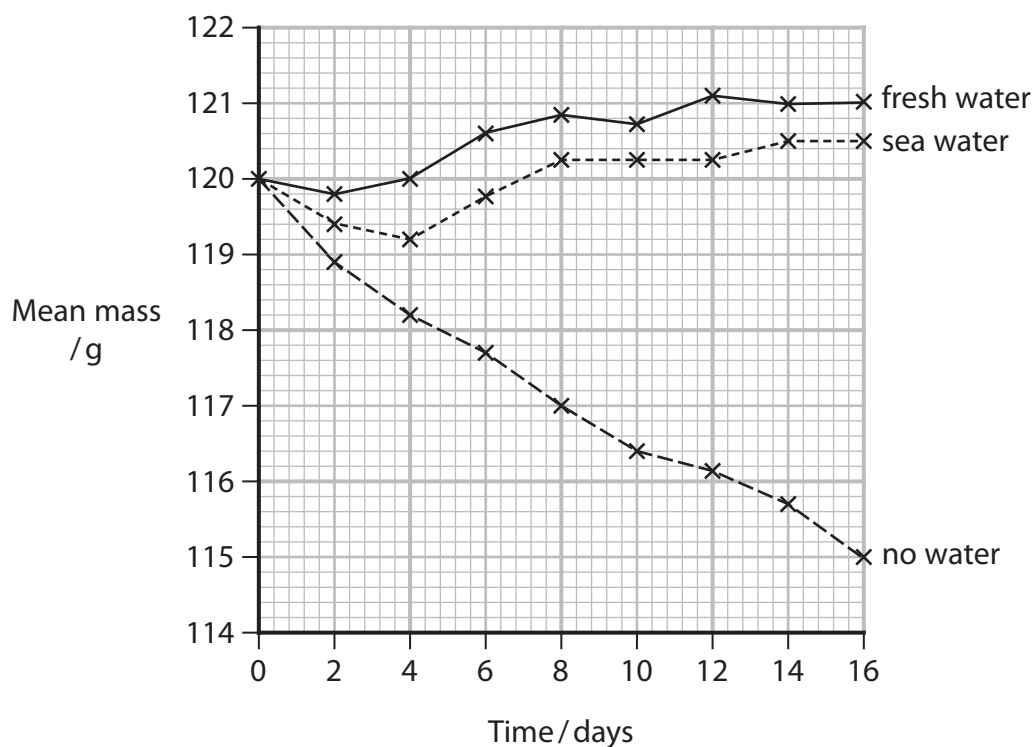
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- (c) The kangaroo rat is another animal that lives in a hot, dry environment.

In an investigation, kangaroo rats were given either fresh water, sea water or no water.

The graph shows the mean body mass of kangaroo rats which were fed on a diet of soya beans over a 16-day period.



Determine the effect of the type of water given on the mean mass of the kangaroo rats in this investigation.

Use the information in the graph to support your answer.

(3)

- (d) The kangaroo rat's kidneys play an important role in the process of conserving water.

Explain how the nephrons (kidney tubules) in the kidneys of the kangaroo rat are able to produce very concentrated urine.

(2)

(Total for Question 5 = 10 marks)